Total No. of Questions : 10]

SEAT No. :

PE3919

[Total No. of Pages : 4

[6582]-236

S.E. (Computer Engineering/Computer Science and Engineering)

PCC-201-COM-CSE : DATA STRUCTURES

(2024 Pattern) (Semester-III)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Use of electronic pocket calculator is allowed.***
5) ***Assume suitable data, if necessary.***

Q1) a) Explain different Asymptotic notations with graph of order of growth. **[6]**

b) Define ADT and Explain Types of Data structures Compare Linear and Non-linear Data Structures. **[8]**

OR

Q2) a) Compare Static and Dynamic Data Structures. Discuss the significance of Persistent and Ephemeral Data Structures. **[6]**

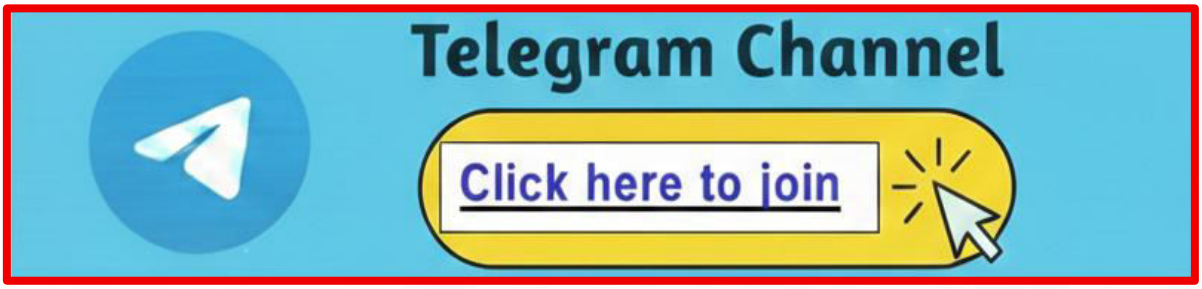
b) Apply step count method to evaluate time complexity of the following python code **[8]**

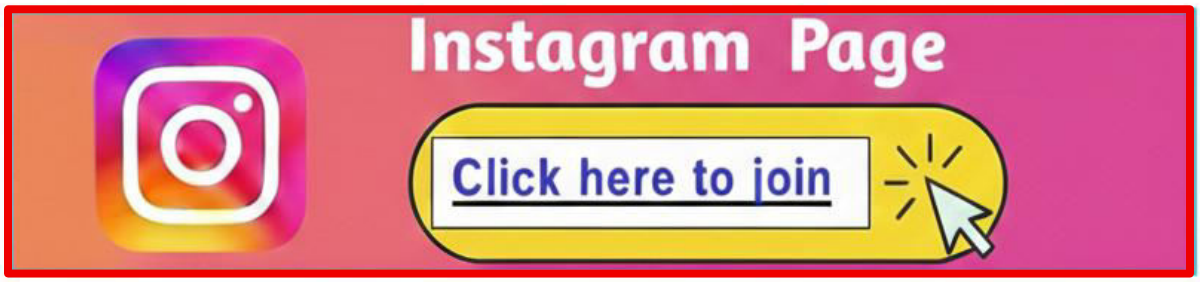
```
def oddEven (n=100):
 for num in range (1, n):
     if num % 2 ==0:
       print (num, 'is Even')
     else:
     print (num, 'is Odd')
```

Justify that the above algorithm is having linear, non-linear, quadratic time complexity. List various Algorithmic Strategies and explain any one with suitable example.

https://t.me/SPPU_TE_BE_COMP

PDF BY VL









© 2026 Vishwajeet Londhe. All Rights Reserved

https://t.me/SPPU_TE_BE_COMP

Q3) a) Define the terms **[6]**

i) Sparse Matrix

ii) Sorting Stability

iii) Internal and External Sort

b) Apply Binary Search algorithm on the following numbers stored in array from A [0] to A [09] as 9, 17, 23, 38, 45, 50, 57, 76, 90, 100 to search numbers 23 and 90. Also note on the time complexity of each search. **[8]**

OR

Q4) a) Define ADT for Arrays and discuss at least 4 operations that are performed on arrays. **[6]**

b) Explain Bubble sort algorithm and sort the given list using bubble sort: 7, 4, 10, 6, 3, 12, 1, 8, 2, 15 Enlist the number of passes required to complete the sort. **[8]**

Q5) a) Explain with neat diagram: **[6]**

i) SLL

ii) DLL

iii) CLL

b) Explain Stack as an ADT and Convert the infix expressions (A+B)*D+E/(F+A*D)+C into postfix representation using stack, Show the status after every step in tabular form. **[8]**

OR

https://t.me/SPPU_TE_BE_COMP

Q6) a) Define and Draw diagram for **[6]**

i) Queue

ii) Circular Queue

iii) Dequeue

b) Write a pseudo code for Stack implementation using Linked List for the following operations **[8]**

i) Create

ii) Push

iii) Pop

iv) Display

Q7) a) Explain characteristics of Good Hash function and Explain any four types of Hash Functions. **[6]**

b) Represent the following key in memory using Linear probing without replacement. Use modulo 10 as your hashing function. 24, 13, 16, 15, 19, 20, 22, 14, 17, 26 Estimate average search time for any element in Hash Table. **[8]**

OR

Q8) a) Write a short note on any three hashing techniques. **[6]**

b) Given the Inputs: 4371, 1323, 6173, 4199, 4344, 9679, 1989} Hash table size = 10 and Hash function = key mod 10. Show the results for the following: **[8]**

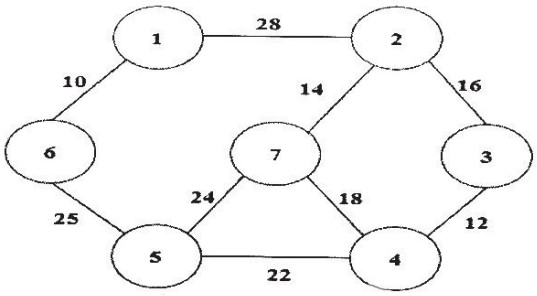
i) Open addressing hash table using Linear Probing

ii) Open addressing hash table using Quadratic Probing

https://t.me/SPPU_TE_BE_COMP

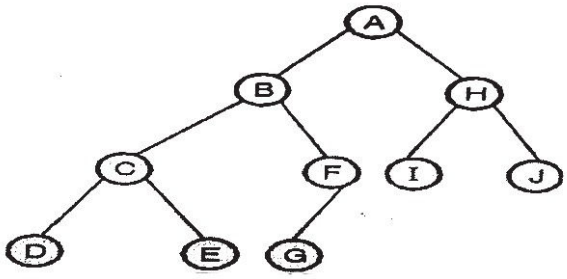
Q9) a) Compare Tree and Graph, Enlist at least two real time applications of each. **[6]**

b) Write Prime's algorithm and apply it on given graph to find Minimum Spanning Tree. Estimate the cost of MST **[8]**

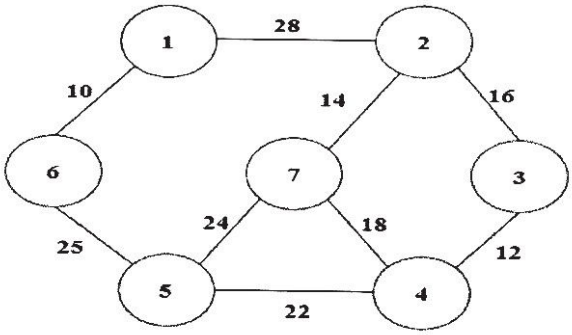


OR

Q10)a) Apply Tree traversal techniques and find in-order, pre-order and post-order traversal sequence of given binary tree. **[6]**



b) Write Kruskal's algorithm and apply it on given graph to find Minimum Spanning Tree. Estimate the cost of MST **[8]**



https://t.me/SPPU_TE_BE_COMP

Total No. of Questions : 10]

SEAT No. :

PE-5720

[Total No. of Pages : 3

[6582]-237

S.E. (Computer / Computer Science Engineering)

PCC - 202 - COM : Object Oriented Programming & Computer Graphics

(2024 Pattern) (Semester - III)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q3. or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.***
2) ***Figures to the right indicate full marks.***
3) ***Neat diagrams must be drawn wherever necessary.***
4) ***Make suitable assumption whenever necessary.***

Q1) a) What are classes and objects? Explain their relationship with the help of a real-world example. **[5]**

b) What is JVM? Explain how java achieves platform independence using JVM and Bytecode. **[5]**

c) Write and explain a Java program using: **[4]**

- If-else decision structure
- for or while loop

OR

Q2) a) What are data members and methods? Describe how they are used in Java to define an object's characteristics and behavior. **[5]**

b) What are Java Class Libraries? Explain how import statements help in accessing predefined classes. **[5]**

c) What are programming paradigms? Describe the four main programming paradigms with suitable examples. **[4]**

Q3) a) What is a constructor? Explain parameterized constructor and default constructor with examples. **[5]**

b) What are arrays? Demonstrate declaration, creation, and initialization of a 1-D and 2-D array with examples. Also explain passing arrays to methods. **[5]**

c) Define a class in Java. Write a Java class with field declaration, method declaration, and method definition. **[4]**

OR

https://t.me/SPPU_TE_BE_COMP

Q4) a) What is method overloading? Write a Java program that overloads a method to calculate the area of a square and rectangle. **[5]**

b) What are set (mutator) and get (accessor) methods? Write an example demonstrating their use. **[5]**

c) What are Java API packages? Explain the use of the import statement with an example. **[4]**

Q5) a) What is inheritance? Explain superclass and subclass with a Java example. **[5]**

b) What is multiple catch? Write a Java program demonstrating handling of different exception types using mutliple catch blocks. **[5]**

c) Explain runtime polymorphism with a suitable Java example demonstrating method overriding. **[4]**

OR

Q6) a) What is protected access specifier? How does it affect accessibility in inheritance? **[5]**

b) Explain try, catch, throw, throws and finally keywords with example code? **[5]**

c) What are abstract classes and abstract methods? How are they different from interfaces? **[4]**

Q7) a) What are the applications of computer graphics in various fields? Explain with examples. **[5]**

b) Write Bresenham circle drawing algorithm. Also explain mathematical foundation of it. **[5]**

c) What is the inside-outside test? How is it used in polygon processing? **[4]**

OR

Q8) a) Explain Scan Line Polygon Fill algorithm. Compare it with Flood Fill and Seed Fill methods. **[5]**

b) Explain DDA line drawing algorithm. Consider line segment from A (-2, -1) to B (6, 3) use DDA line drawing algorithm to rasterize this line **[5]**

c) Explain the following term with example: **[4]**

i) Windowing

ii) Clipping

https://t.me/SPPU_TE_BE_COMP

Q9) a) Define 2D transformations. Explain translation, scaling, rotation, and shear with their matrix representations. **[5]**

b) Describe rotation about an arbitrary axis in 3D. Explain the steps involved. **[5]**

c) What is Oblique projection? Explain Cavilier and Cabinet projection. **[4]**

OR

Q10) a) What is orthographic projection? Explain Isometric, Diametric, and Trimetric projections with suitable examples. **[5]**

b) Explain 3D translation, scaling, rotation, and shear with their matrix forms. **[5]**

c) Perform x shear on a triangle having A (2, 1), B (4, 3), C (2, 3) consider shx=2. **[4]**



https://t.me/SPPU_TE_BE_COMP

Total No. of Questions : 10]

SEAT No. :

PE-5721

[Total No. of Pages : 3

[6582]-238

S.E. (Computer/Computer Science& Engg.)

PCC-203-COM : OPERATING SYSTEMS

(2024 Pattern) (Semester - III)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates :

1) ***Attempt Q.1, or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8 and Q.9 or Q.10.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Numeric figures in square brackets [] indicate full marks.***
4) ***Use of electronic pocket calculator is allowed.***
5) ***Assume suitable data, if necessary.***

Q1) a) List different types of Operating Systems? Describe any two of them. **[5]**

b) What is a system call? Explain its need in operating systems. **[5]**

c) Describe the various functions of an Operating System. **[4]**

OR

Q2) a) Explain the layered approach of OS design with the help of diagram. Discuss its advantages and disadvantages. **[5]**

b) Explain the main objectives of an Operating System. How does an OS act as a resource manager and control program? Give examples. **[5]**

c) Explain Microkernel Operating Systems? **[4]**

Q3) a) Explain Process States and Transitions diagram? **[7]**

b) Explain the following types of Schedulers. **[7]**

i) Short Term

ii) Long Term

iii) Medium Term

OR

P.T.O.

https://t.me/SPPU_TE_BE_COMP

Q4) a) Explain the Process Control block in detail with the help of a diagram. **[7]**

b) Draw Gantt chart and calculate Avg. turnaround time, Avg. waiting time for the following process using SJF non-pre-emptive and round robin with time quantum 0.5 Unit. **[7]**

Process	Burst time	Arrival Time
P1	2	10
P2	1	10
P3	1	11
P4	1	12

Q5) a) Write and explain Deadlock Avoidance Bankers Algorithm. **[6]**

b) Explain necessary conditions for occurrence of deadlock. **[4]**

c) What is semaphore? Justify how semaphore is used to solve critical section problems. **[4]**

OR

Q6) a) Write a short note on the following with an example? **[6]**

i) Semaphore

ii) Monitors

iii) Mutex

b) Explain Deadlock prevention and deadlock recovery with example? **[4]**

c) What is semaphore? Write a solution to the Reader Writer problem using Semaphore with Readers as priority. **[4]**

Q7) a) Explain FIFO and LRU page replacement policies? Consider page sequences 2, 3, 2, 1, 5, 4, 5, 3, 2, 2,2, 5, 2 and discuss working with FIFO replacement policy. Also count page faults. (Use no. of frames = 3). **[7]**

b) Write a short note on following with diagram. **[7]**

i) Segmentation

ii) Demand Paging

OR

Q8) a) Explain following placement strategies: First Fit, Best Fit and Worst Fit. **[7]**

b) What is internal fragmentation? Explain with a suitable diagram and example. **[7]**

https://t.me/SPPU_TE_BE_COMP

Q9) a) Explain File operation and Directory structure? **[5]**

b) Explain File Allocation Methods? **[5]**

c) Compare FAT, NTFS, EXT, and HFS file systems based on structure, performance, and security. **[4]**

OR

*Q10)*a) Describe Boot-block and Bad blocks? **[5]**

b) Explain Disk formatting? **[5]**

c) Explain LOOK scheduling. How is it different from SCAN? **[4]**



https://t.me/SPPU_TE_BE_COMP

Total No. of Questions : 10]

SEAT No. :

PE-5722

[Total No. of Pages : 3

[6582]-239

S.E. (Computer/Computer Science & Engineering)

MDM-221 : DIGITAL ELECTRONICS & LOGIC DESIGN

(2024 Pattern) (Semester - III)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates :

1) *Attempt Q.1, or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8 and Q.9 or Q.10.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Numeric figures in square brackets [] indicate full marks.*
4) *Use of electronic pocket calculator is allowed.*
5) *Assume suitable data, if necessary.*

Q1) a) Simplify and implement following expression using K-Map **[5]**

$Y = \sum m\ (0, 1, 2, 5, 7, 8, 9, 10, 13, 15)$

b) Minimise the following expression and implement using logic gates. **[5]**

$(Y = (A + B + \overline{C})\ (A + \overline{B} + \overline{C})\ (\overline{A} + B + C)\ (\overline{A} + \overline{B} + \overline{C})\ (A + B + C)$

c) Perform the subtraction using 1's Complement and 2's Complement method $(151)_{10} - (76)_{10}$. **[4]**

OR

Q2) a) Derive the POS of the following expression using K-Map and implement using logic gates. **[5]**

$F(A, B, C, D) = \prod M\ (0, 2, 3, 8, 9, 12, 13, 15)$

b) Using Boolean algebra theorems, simplify the following expression and implement using logic gates. **[5]**

$F = (A + B)\ (A + C)\ (\overline{A} + \overline{B})$

c) Differentiate between Signed Magnitude, 1's Complement, and 2's Complement representation. Represent –23 using all three methods. **[4]**

P.T.O.

https://t.me/SPPU_TE_BE_COMP

Q3) a) Explain the working of a 4-bit Binary Adder using IC-7483. Write its features and applications. **[5]**

b) Explain the working of a BCD-to-Excess-3 code converter with truth table and circuit diagram. **[5]**

c) State the difference between half and full adder. **[4]**

OR

Q4) a) Draw and explain 4 bit BCD subtractor using IC-7483. Explain the operation on BCD number 5-7. **[7]**

b) Implement the following Boolean function using 8 : 1 multiplexer **[7]**

$$(\bar{A}\ B\ D) + (\bar{A}\ B\ \bar{D}) + (A\ C\ D) + (\bar{B}\ C\ D) + (\bar{A}\ \bar{C}\ D)$$

Q5) a) Convert the following flip-flops : **[5]**

i) SRtoJK.

ii) JK to T

b) Draw and explain 4-bit asynchronous up-counter using JK flip flop. Also draw the necessary timing diagram. **[5]**

c) With neat diagrams explain the working of the following types of shift registers **[4]**

i) Serial-in, serial-out

ii) Parallel-in, serial-out

OR

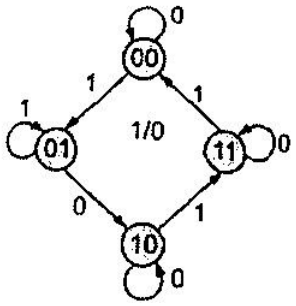
Q6) a) Define the term "Excitation Table". Write the excitation tables for SR, JK, D, and T flip-flops and explain their importance. **[5]**

b) Design a Mod-6 asynchronous counter using T flipflop. **[5]**

c) Differentiate between Combinational Circuit and Sequential Circuit. **[4]**

https://t.me/SPPU_TE_BE_COMP

Q7) a) Design a synchronous sequential circuit using JK flipflop for the given state diagram. **[5]**



b) Write a short note on characteristics of logic families. **[5]**

c) Explain the differences between a Finite State Machine (FSM) and an Algorithmic State Machine (ASM) with examples. **[4]**

OR

Q8) a) Explain in detail Mealy and Moore machine **[5]**

i) State diagram and

ii) State table

b) Draw an ASM chart and state table for 2 bit up down counter having mode control input : **[5]**

M=1 : Up counting

M=0 : Down counting

The circuit should generate a output 1 whenever the count becomes minimum or maximum

c) Compare Mealy Machine with Moore Machine **[4]**

Q9) a) Draw and explain the architecture of PLA. **[7]**

b) Explain the difference between PLA, PAL, and ROM in terms of programmability and flexibility. **[7]**

OR

*Q10)*a) Design a Programmable Logic Array (PLA) to implement the following functions : **[5]**

i) $F1 = \overline{A}B + A\overline{B} + C$

ii) $F2 = \overline{A}\,\overline{B}C + AB$

b) Design BCD to excess-3 code converter and implement it using PAL. **[5]**

c) Write short note on different types of PLD' s. **[4]**



https://t.me/SPPU_TE_BE_COMP

Total No. of Questions :8]

SEAT No. :

PE3920

[Total No. of Pages : 2

[6582]-240

S.E. (Comp. Engg.)/(Comp. Sc.& Engg.)/AI & DS/ Comp. Sc.& Engg. (AI)/Comp. Sc.& Engg. (D.S.)/Comp. Sc./ Comp. Sc.& Design)

VEC-232 : UNIVERSAL HUMAN VALUES AND PROFESSIONAL ETHICS

(2024 Pattern) (Semester-III)

Time : 2 Hours] *[Max. Marks : 35*

Instructions to the candidates:

1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Numeric figures in square brackets [] indicate full marks.*
4) *Use of electronic pocket calculator is allowed.*
5) *Assume suitable data, if necessary.*

Q1) a) What do you mean by 'Value Education'? Explain the basic guidelines for Value Education. **[5]**

b) What is the Basic Human Aspiration? Is it the same for everyone? **[3]**

OR

Q2) a) Why is right understanding required for ensuring physical facility? How does it result in mutual prosperity? Illustrate with the help of two personal examples. **[5]**

b) Explain the process of Self-exploration with the help of a diagram. **[3]**

Q3) a) Human being is the co-existence of the Self and the Body. What are the three specific distinguishing attributes of the Self and the Body? Explain with examples. **[5]**

b) Body is the instrument of the Self. Explain. **[4]**

OR

Q4) a) What is the purpose of the Self? What is the purpose of the Body? What would be the right utilization of the Body? **[5]**

b) Distinguish between intention and desire. **[4]**

https://t.me/SPPU_TE_BE_COMP

Q5) a) Give one example each of over evaluation, under evaluation and otherwise evaluation. Why are these evaluations leading to disrespect? **[5]**

b) Differentiate between Glory and Gratitude. **[4]**

OR

Q6) a) Distinguish between intention and competence. What is the outcome when we confuse between the two and we doubt the intention of the other? **[5]**

b) What is the role of physical facility in the fulfilment of relationship? **[4]**

Q7) a) Explain the activities in the four orders of nature. How are the activities in the human order qualitatively different from those of other three orders? **[5]**

b) What are the Professional Ethics? **[4]**

OR

Q8) a) What do you understand by the terms 'holistic development' and 'universal human order'? **[5]**

b) Explain Ethical Human Conduct. Discuss its characteristics. **[4]**

● ● ●

https://t.me/SPPU_TE_BE_COMP